

Linux RAID 5 Lab Note

Create, break and repair a Redundant Storage Array.

Here is how I have done it!

I have started with a locked Google Chromebook. It didn't permit me to do the operations.

So, I chose google cloud, which is fantastic for the learning environment, and it is secure.

Part 1: Preparing the Virtual Disks

Use the following commands to create “fake” hard drives out of files.

1. Create 3 empty disks: `fallocate -l 100M disk1.img disk2.img disk3.img`
2. Attach Disk 1 to the system: `sudo losetup -fP disk1.img`
3. Attach Disk 2 to the system: `sudo losetup -fP disk2.img`
4. Attach Disk 3 to the system: `sudo losetup -fP disk3.img` [here letter 'p' should be capital]
5. Confirm all their names (usually loop0,1,2): `losetup -a`

Part 2: Building the Raid

Now, we need to combine the disks and prepare them for files.

6. Install the RAID tool: `sudo apt update && sudo apt install mdadm -y`
7. Create the RAID 5 Array: `sudo mdadm --create --verbose /dev/md0 --level=5 --raid-devices=3 /dev/loop0 /dev/loop1 /dev/loop2`
8. Now format the Array: `sudo mkfs.ext4 /dev/md0`
9. Next, create a folder to hold the files: `sudo mkdir -p /mnt/raid_lab`
10. Then, Plug in the RAID to that folder: `sudo mount /dev/md0 /mnt/raid_lab`

Part 3: Testing and Resilience

To check that the system works:

11. Create a secret file on the RAID: `echo "RAID is my secret data!" | sudo tee /mnt/raid_lab/secret.txt`
12. Simulate a disk failure: `sudo mdadm /dev/md0 --fail /dev/loop1`
13. Verify the data still exists: `cat /mnt/raid_lab/secret.txt`
14. Check the 'injured' status: `cat /proc/mdstat`

Part 4: The Repair (Hot-Swap) - Fix the server and return to full health.

15. Remove the broken disk: `sudo mdadm /dev/md0 --remove /dev/loop1`

16. Wipe the disk: `sudo wipefs -a /dev/loop1`

17. Add the new disk back to the array: `sudo mdadm /dev/md0 --add /dev/loop1`

18. Final health check: `sudo mdadm --detail /dev/md0`